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Advanced Methods of Structural Analysis

Second Edition

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Strength, Stability, Vibration

Second Edition

 Springer

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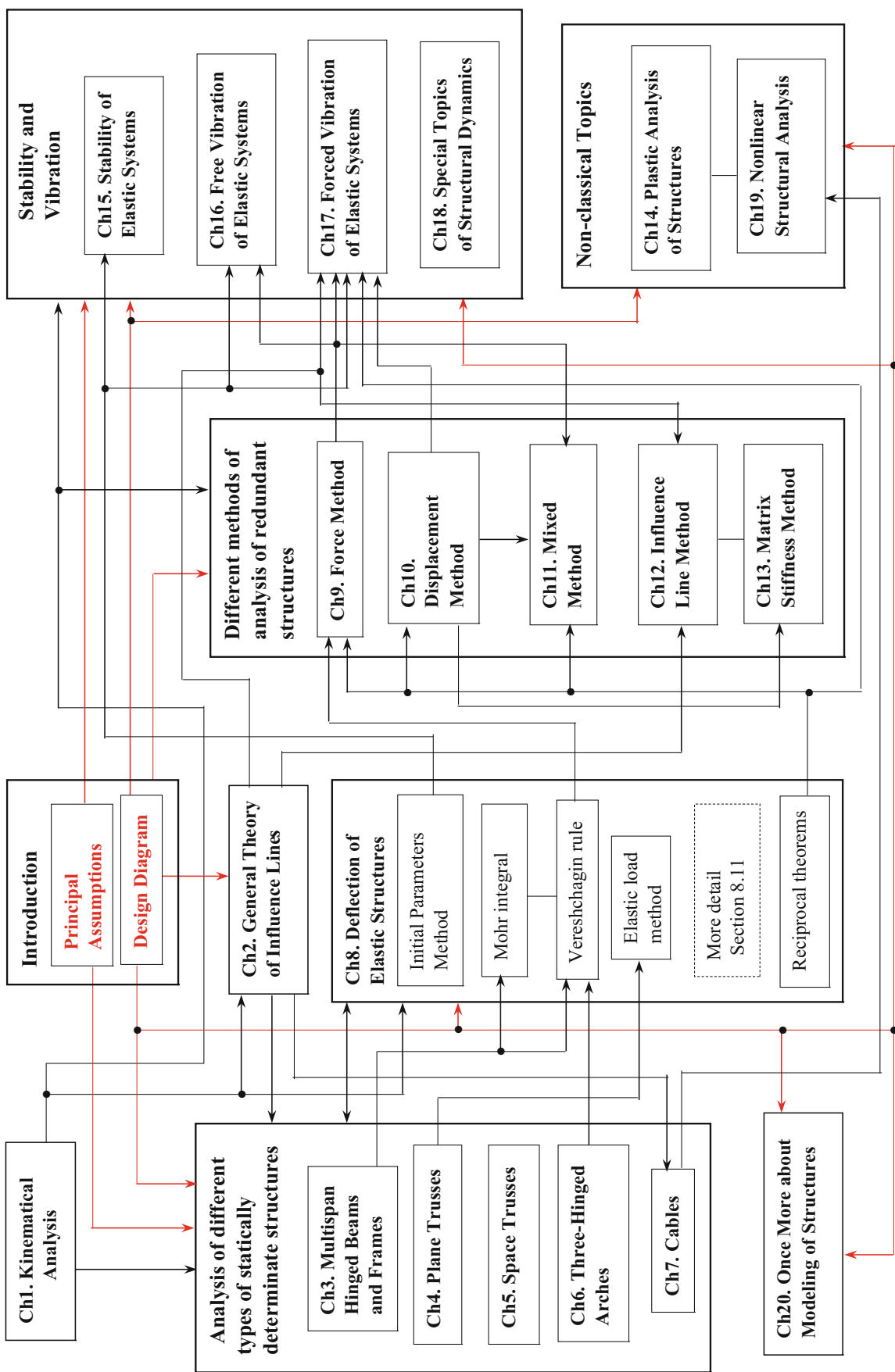
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Book Road Map



Preface to the Second Edition

The first edition of the book entitled “Advanced Methods of Structural Analysis” (593 pages) was published by Springer Publishing House in 2010 (Karnovsky and Lebed 2010). Authors present the second revised and expanded edition of the textbook under the extended name “Advanced Methods of Structural Analysis: Strength, Stability, Vibration.”

The authors retained the general structure of the first edition of the book Advanced Methods of Structural Analysis, but in the second edition of the book key concepts have been added, new chapters have been introduced, and existing chapters have been significantly expanded. These additions relate to the theory, analysis procedures, detailed illustrative examples, and discussion of the results and comments.

All additions pursued the following purposes:

- Draw reader’s attention to the conceptual provisions that go through the whole course of the theory of structural mechanics. Among them are design diagram of a structure, equilibrium conditions for the whole system and any part of it, correspondence of the stress-deformable state of the system to the superimposed constraints, and invariance of the analysis results from the accepted method of analysis of the structure.
- Acquaint the reader with some questions of the theory of structures that are insufficiently covered in the educational literature or even completely absent in them.
- Expand the scope of the classical methods used for analyzing the mechanics of rod systems, and show typical examples of the advantages and disadvantages of analysis methods.
- Prepare the reader for the idea of presentation of the structure by different design diagrams and justification of its final version that meets various but often contradictory requirements.

The following six new chapters have been introduced:

Chapter 5: Space Frameworks. The reader is provided with detailed information about the methods of formation of spatial rod structures, the conditions of attachment to the base, and the method of connecting two 3-D structures. Attached and released framework concepts are introduced. Various types of spatial structures (meshwork structures, compound, complex space frameworks) and procedure for their analysis are discussed. Detailed analysis of the Schwedler’s dome is presented.

Chapter 17: Dynamics of Elastic Systems. Forced Vibration. This chapter is devoted to the dynamic analysis of structures with finite and infinite number of degrees of freedom subjected to force and kinematical excitations. The steady-state and transitional vibrations are considered. Dynamical analysis is carried out in terms inherent to the deformable structures adopted in the classical course of structural mechanics. The continuity of classic analysis methods (such as force and displacement methods, initial parameter method) to the problems of dynamic behavior of deformable systems is presented.

Material covered in new Chapters 5 and 17 is considered within the framework of classical assumptions.

Chapter 18: Special Topics of Structural Dynamics*. The peculiarity of this chapter is that each paragraph of this chapter contains an additional assumption that does not contradict the assumptions adopted in the classical structural analysis. This allows for constructing a refined mathematical model of the object, discovering new qualitative aspects of system behavior, and explaining the paradoxes arising under the classical assumptions.

Chapter 19: Nonlinear Structural Analysis*. A feature of this chapter is that each paragraph of this chapter contains a **rejection** of at least one of the conventional assumptions of the classical structural analysis. The need to consider such problems is that in an arbitrary deformable system inevitably there are various kinds of nonlinearity. Taking them into account allows discovering new effects inherent to a real structure. Analytical solutions of a number of nonlinear problems of strength, stability, and vibrations of beams are presented; these solutions can be considered as test result for an alternate method.

Chapter 20: Once More About Modeling of Structures. This chapter summarizes modeling deformable structures. The chapter contains requirements and factors influencing the choice of design diagram as well as concepts for transition from a constructive scheme of a structure to its design diagram. It also notes an inevitable trade-off between a detailed design

diagram, for which precise results can be obtained, and a simplified diagram, for which less accurate results can be obtained. This chapter also discusses the difference between *analysis* of a structure and *direct design* of a structure.

Chapter 21: Fundamental Developments in the History of Structural Mechanics. This chapter covers only two stages—the period of the formation of the science of strength (1638–1826) and the creation of the classical theory of structures itself (from 1826). The following set of issues are discussed: the objective reasons behind deadlock state of the limit state method, the essence of the Claude-Louis Navier reform (1826) and the need to switch from the limit state method to the working state method, the difference between these methods, and the disadvantages and merits of structural analysis utilizing the working state method.

The following additions have been made:

Introduction is substantially revised and supplemented. The subject of detailed discussion is the concept of design diagram, which is the cornerstone of structural mechanics course, and determines the behavior of the system and the method of its analysis.

Chapter 8: A number of classical methods for determining displacements are added (a fictitious beam method, energy methods, the principle of the smallest work, etc.).

Chapter 9: Classical methods of beam analysis are added (three moment equations and focal ratio method).

Chapter 16: A new material related to the application of the method of initial parameters (16.4.4–16.4.5) and the method of displacements (16.4.6) in problems of dynamics is added.

Chapters 2–4: Construction of influence lines by kinematical method for beams and plane trusses is added.

Also, examples are added in separate chapters; reference material in the Appendix and the list of bibliography are expanded.

Much attention is paid to verifying the results obtained by applying different methods of analysis to the same design diagram. Solving problems of a certain class using various methods allows the reader to compare their effectiveness and evaluate their strengths and weaknesses. This approach is of great methodological significance, since it allows formulating recommendations with an indication of the limits of the effective application of methods.

Authors will appreciate comments and suggestions to improve the current edition. All constructive criticism will be accepted with gratitude.

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Preface to the First Edition

Theory of the engineering structures is a fundamental science. Statements and methods of this science are widely used in different fields of engineering. Among them are the civil engineering, shipbuilding, aircraft, robotics, space structures, as well as numerous structures of special types and purposes—bridges, towers, etc. In recent years, even micromechanical devices have become objects of structural analysis.

Theory of the engineering structures is alive and is a very vigorous science. This theory offers an engineer-designer a vast collection of classical methods of analysis of various types of structures. These methods contain in-depth fundamental ideas and, at the present time, they are developed with sufficient completeness and commonness, aligned in a well-composed system of conceptions, procedures, and algorithms; use modern mathematical techniques; and are brought to elegant simplicity and perfection.

We now live in a computerized world. A role and influence of modern engineering software for the analysis of structures cannot be overestimated. The modern computer programs allow providing different types of analysis for any sophisticated structure. As this takes place, what is the role of classical theory of structures with its in-depth ideas, prominent conceptions, methods, theorems, and principles? Knowing classical methods of structural analysis is necessary for any practical engineer. An engineer cannot rely only on the results provided by a computer. Computer is a great help in modeling different situations and speeding up the process of calculations, but it is the sole responsibility of an engineer to check the results obtained by a computer. If users of computer engineering software do not have sufficient knowledge of fundamentals of structural analysis and understanding of physical theories and principal properties of structures, then he/she cannot check obtained numerical results and their correspondence to an adopted design diagram, as well as explain results obtained by a computer. Computer programs “. . . can make a good engineer better, but it can make a poor engineer more dangerous” [Cook R.D., Malkus D.S., Plesha M. E (1989) Concepts and applications of finite element analysis, 3rd edn. Wiley, New York]. Only the knowledge of fundamental theory of structures allows to estimate and analyze numerical data obtained from a computer; predict the behavior of a structure as a result of changing a design diagram and parameters; design a structure which satisfies certain requirements; perform serious scientific analysis; and make valid theoretical generalizations. No matter how sophisticated the structural model is, no matter how effective the numerical algorithms are, no matter how powerful the computers are that implement these algorithms, it is the engineer who analyzes the end result produced from these algorithms. Only an individual who has a deep knowledge and understanding of the structural model and analysis techniques can produce a qualitative analysis.

In 1970s, one of the authors of this book was a professor at a structural engineering university in Ukraine. At that time, computers were started to be implemented in all fields of science, structural analysis being one of them. We, the professors and instructors, were facing a serious methodical dilemma: Given the new technologies, how to properly teach the students? Would we first give students a strong basis in classical structural analysis and then introduce them to the related software, or would we directly dive into the software after giving the students a relatively insignificant introduction to classical analysis? We did not know an optimal way to solve this problem. On this subject we have conducted seminars and discussions on a regular basis. We have used these two main teaching models, and many different variations of them. The result was somewhat surprising. The students that were first given a strong foundation in structural analysis quickly learned how to use the computer software, and were able to give a good qualitative analysis of the results. The students that were given a brief introduction to structural analysis and a strong emphasis on the computer software at the end were not able to provide qualitative results of the analysis. The interesting thing is that the students themselves were criticizing the latter teaching strategy.

Therefore, our vision of teaching of structural analysis is as follows: in the first step, it is necessary to learn analytical methods, perform detailed analysis of different structures *by hand* in order to feel the behavior of structures, and correlate their behavior with obtained results; the second step is a computer application of engineering software.

Authors wrote the book on the basis of their many years of experience of teaching the structural analysis at the universities for graduate and postgraduate students as well as on the basis of their experience in consulting companies.

This book is written for students of universities and colleges pursuing civil or structural engineering programs, instructors of structural analysis, and engineers and designers of different structures of modern engineering.

The objective of the book is to help a reader to develop an understanding of the ideas and methods of structural analysis and to teach a reader to estimate and explain numerical results obtained by hand; this is a fundamental stone for preparation of readers for numerical analysis of structures and for use of engineering software with full understanding.

The textbook offers the reader the fundamental theoretical concepts of structural analysis, classical analytical methods, algorithms of their application, comparison of different methods, and a vast collection of distinctive problems with their detailed solution, explanation, analysis, and discussion of results; many of the problems have a complex character. Considered examples demonstrate features of structures, their behavior, and peculiarities of applied methods. Solution of all the problems is brought to a final formula or number.

Analyses of the following structures are considered: statically determinate and indeterminate multi-span beams, arches, trusses, and frames. These structures are subjected to fixed and moving loads, changes of temperature, settlement of supports, and errors of fabrication. Also, the cables are considered in detail.

In many cases, same structure under different external actions is analyzed. It allows the reader to be concentrated on one design diagram and perform complex analysis of behavior of a structure.

In many cases, same structure is analyzed by different methods or by one method in different forms (for example, displacement method in canonical and matrix forms). It allows to perform comparison analysis of applied methods and see advantages and disadvantages of different methods.

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This chapter is devoted to the analysis of cables under fixed loads of different types. Among them are concentrated loads, and uniformly distributed load along the horizontal line and along the cable itself. Important formulas for analysis of the cable subjected to arbitrary loads are derived; they allow determining the *changing* of reactions, internal forces, and shape due to any *additional* live loads. Relationships between the thrust, internal forces, and total length of a cable are established. The influence of elastic properties of a cable is discussed.

7.1 Preliminary Remarks

The cables as the permanent members of the load-bearing structures are used extensively in modern engineering. Some examples of cabled structures are suspension bridges (Fig. 7.1), anchoring systems of different objects such as guy-rope of the masts, sea drilling platforms, stadium covering, cableways, floating breakwaters, etc. Cables are also used as the temporary guys during erection of the structures.

Cables are made from high-strength steel wires twisted together, and present a flexible system, which can resist only axial tension. The cables allow covering very large spans; modern suspension bridges permit coverage of spans hundreds of meters in length. This may be explained by two reasons. (1) In axial tension, the stresses are distributed uniformly within all areas of cross section, so the material of a cable is utilized in full measure; (2) cables are made from steel wires with very high ultimate tensile strength ($\sigma_u \cong 1860 \text{ MPa}$, while for structural steel ASTM-A36 $\sigma_u \cong 400 \text{ MPa}$). Therefore the own weight of a load-bearing structure becomes relatively small and the effectiveness of application of cables increases with the increasing of the spans.

A cable as a load-bearing structure has several features. One of them is the vertical loads that cause horizontal reaction, which, as in case of an arch, is called a thrust. However, the behavior of the cable and the arch is fundamentally different. Bending moments, shear, and axial *compressive* forces arise in the arch, and *only tensile axial* forces in the cable.

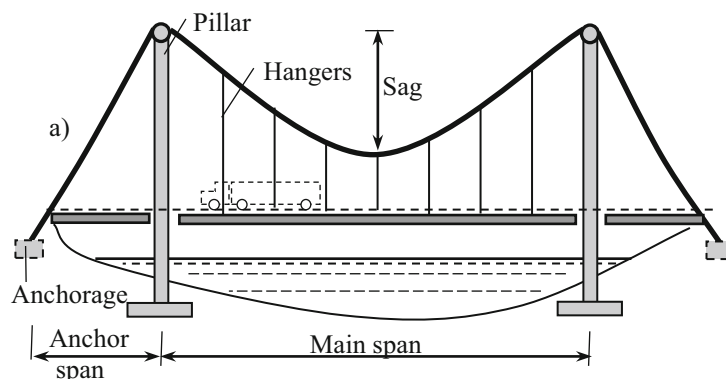


Fig. 7.1 Suspension bridge

Another feature of cables is the high sensitivity of their shape, depending on the total length of the cable, type of the load, as well as a load location along the span. In other words, a cable is a *flexible* system. The fundamental feature of a cable is its unknown shape in advance. Defining the shape of a cable is one of the important problems.

The following types of cables are considered:

1. Cable with self-weight ignored (a cable subjected to concentrated loads, to uniformly distributed load, as well as to arbitrary dead load and after that additional any live load)
2. Cable with self-weight (with or without external load)

Under these loads the cable shape takes different forms. Shape of a cable in case of concentrated loads only is defined by slopes of straight portions of a cable. Shapes of a cable in case of distributed load along horizontal and along cable itself are parabola and catenary, respectively. Note that the intensity of the load due to weight of cable itself along its axis is constant, while the intensity of the same weight along horizontal projection of the cable is variable.

Assumptions:

1. The cable is an inextensible one (elastic properties of the cable will be considered in Sect. 7.6).
2. The cable is perfectly flexible; that is, the cable does not resist to shear and bending. In this case the internal force in any point of a cable is tensile, which is directed along the tangent at this point of a cable.

Horizontal component of a tension force is called a thrust. It will be shown later that thrust remains constant along the length of the cable.

7.1.1 Direct and Inverse Problems

The simplest problem is finding thrust, internal forces, and total length of a cable if the shape of a cable is given. Solution of this problem is well known. However, for practice this type of problem does not have a reasonable sense because shape of the cable cannot be known in advance. Instead, an engineer knows allowable horizontal force, which a tower can resist at the top. Based on this, we will consider the following two fundamental practical problems for cables under fixed loads:

1. Find the shape of a cable and internal forces, if a thrust is given. This type of problem is called the thrust-shape problem.
2. Compute thrust and internal forces, if a total length of a cable is given. This type of problem is called the length-thrust problem.

7.1.2 Fundamental Relationships

Design diagram of a cable with supports without their mutual displacements is shown in Fig. 7.2a. The cable is loaded by some vertical concentrated loads P_i (the distance from left support is x_i) and arbitrary load q distributed along horizontal axis. Since the cable is perfectly flexible the axial force N_x at any section is directed along tangent at this section. Axial forces N_A and N_B at the supports A and B are resolved into two directions. They are the vertical direction and direction $A-B$. These forces are denoted as r_A and h_A for support A and r_B and h_B for support B . Summing of all forces on the horizontal axis leads to the relationship $h_A = h_B = h$.

The vertical component of the reaction at B :

$$r_B \rightarrow \sum M_A = 0 : r_B l - \sum P_i x_i - \int_l q(x) x dx = 0$$

$$r_B = \frac{1}{l} \left[\sum P_i x_i + \int_l q(x) x dx \right] \quad (7.1)$$

This expression coincides with general formula for reaction of simply supported beam. Therefore if axial forces at supports are resolved as shown in Fig. 7.2a, then a vertical component of axial force at support equals to corresponding reaction of the reference beam.

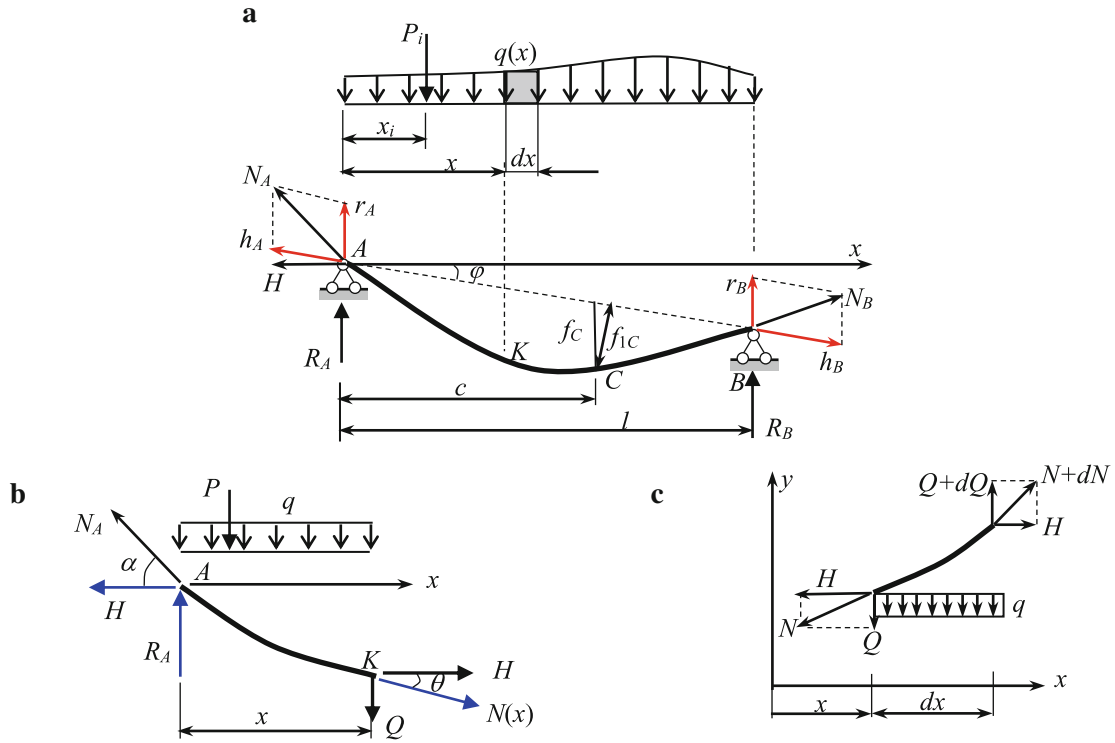


Fig. 7.2 (a) Cable subjected to arbitrary load; (b) free-body diagram of the part of the cable; (c) free-body diagram of the elementary portion of the cable

Now we can calculate ordinate of the cable at section C with abscissa c . Since the bending moment at any section C is zero,

$$M_C = r_A \cdot c - h_A \cdot f_{1C} - \sum P_i(c - x_i) - \int_0^c q(x)(c - x)dx = 0 \quad (7.2)$$

$$f_{1C} = \frac{1}{h} \left[r_A \cdot c - \sum P_i(c - x_i) - \int_0^c q(x)(c - x)dx \right]$$

where f_{1C} is perpendicular to the inclined chord AB .

It is obvious that the expression in bracket (7.2) presents the bending moment at section C of the reference beam. Therefore

$$f_{1C} = \frac{M_C^0}{h} \quad (7.3)$$

To determine the vertical coordinate of the point C we need to modify this formula. For this we need to introduce a concept of a thrust H , which is a horizontal component of the axial force at any section. From Fig. 7.2b we can see that equilibrium equation $\sum X = 0$ leads to the formula $H = \text{const}$ for any point of the cable subjected to arbitrary loads. At any support $h = H / \cos \varphi$. Since $f_{1C} = f_C \cos \varphi$, the expression (7.3) can be rewritten as $f_C \cos \varphi = M_C^0 / (H / \cos \varphi)$, which leads to the fundamental formula

$$f_C = \frac{M_C^0}{H} \quad (7.4)$$

where f_C is measured vertically from the inclined chord AB . This parameter is called the sag of the cable.

Formula (7.4) defines the shape of the cable subjected to any vertical load. The sag is proportional to the bending moment of the “reference beam”; proportional coefficient is $1/H$. The sag and y -coordinate are equal if supports are located on the same elevation. Now the total vertical reactions of supports may be determined in terms of thrust as follows:

$$\begin{aligned} R_A &= r_A + h \sin \varphi = r_A + H \tan \varphi \\ R_B &= r_B - h \sin \varphi = r_B - H \tan \varphi \end{aligned} \quad (7.5)$$

The concept of “shear” at any section of the reference beam will be helpful. The portion length of x is loaded by uniformly distributed load q , concentrated force P , reaction R_A , thrust H , and axial force $N(x)$ as shown in Fig. 7.2b. The beam shear is $Q(x) = R_A - P - qx$. At section x (point K) three concurrent forces act. They are $Q(x)$, $N(x)$, and H . It is obvious that for any section

$$N(x) = \sqrt{H^2 + Q^2(x)} \quad (7.6)$$

Now let us derive the differential relationships between load q and ordinate of the cable y . The angle θ between the axial line of a cable and x -axis (Fig. 7.2b) obeys the equation

$$\tan \theta = \frac{dy}{dx} = \frac{Q}{H} \quad (7.7)$$

To derive a relationship between Q and intensity of the load q , consider a free-body diagram of the elementary portion of the cable (Fig. 7.2c). The axial forces at the left and right ends of a portion are N and $N + dN$, respectively. Their components are H and Q at the left end, while at the right end H and $Q + dQ$.

Equilibrium equation $\sum Y = 0$ leads to the following relationship: $dQ/dx = q$. Taking into account (7.7), the required relationship between y and q becomes

$$\frac{d^2y}{dx^2} = \frac{q(x)}{H} \quad (7.8)$$

The first and second integrating of Eq. (7.8) leads to the expressions for slope and y -ordinate, respectively. The constant of integration should be determined from the boundary conditions.

The determination of stresses in cables in case of their various loadings may be facilitated using the influence line method. We return to relation (7.4). Suppose that the supports are at the same level. It is obvious that in the case of loading by concentrated force in the middle of the span, the greatest deflection f_C is at point C under the force. We can transform the expression for the thrust H to the equation of influence line for the thrust: $IL(H) = \frac{IL(M_C^0)}{f_C}$.

Influence line of the bending moment at the middle point C of the substitute beam has the form of a triangle with a vertex at the point C ; the ordinate of the vertex is $l/4$. Therefore, the influence line of the thrust has the same form with the ordinate $l/4f$, where f is the cable sag in the middle.

Let the cable be loaded with a uniformly distributed load q along the entire span. In this case, the thrust H becomes

$$H_l = q\omega_l = q \cdot 2 \cdot \frac{1}{2} \cdot \frac{l}{2} \cdot \frac{l}{4f} = \frac{ql^2}{8f}$$

If the load occupies the left (right) half of the span, then the thrust is

$$H_{l/2} = q\omega_{l/2} = q \cdot \frac{1}{2} \cdot \frac{l}{2} \cdot \frac{l}{4f} = \frac{ql^2}{16f}$$

7.2 Cable with Neglected Self-Weight

This section contains analysis of the perfectly flexible inextensible cables subjected to concentrated loads, as well as distributed loads along horizontal axis through the entire span. For both loading (concentrated and distributed) of a cable the thrust-shape and length-thrust problems are considered. Often, these problems are referred as direct and inverse problems.

7.2.1 Cables Subjected to Concentrated Load

In this case of the loading each portion of the cable between two adjacent forces presents the straight segment. The simplest design diagram of a cable, subjected to one concentrated load, is presented in Fig. 7.3a. Parameters of the system are $a = 10$ m, $l = 25$ m, and $P = 20$ kN.

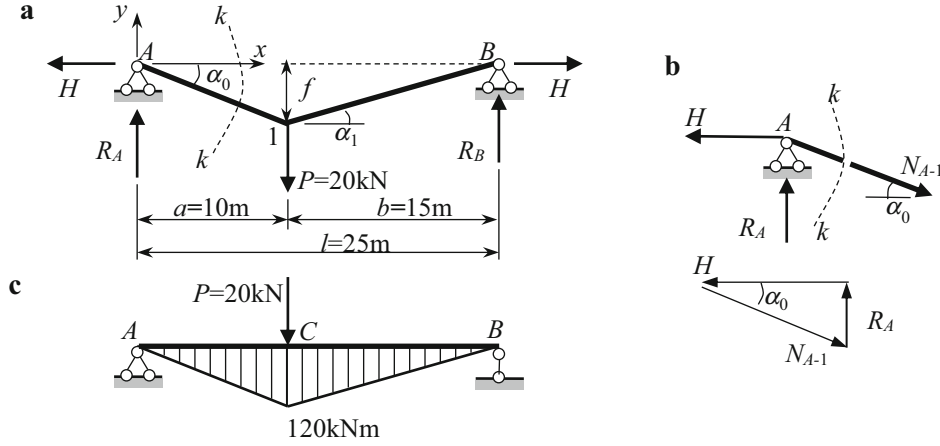


Fig. 7.3 (a) Design diagram of the cable; (b) free-body diagram; (c) reference beam and corresponding bending moment diagram

7.2.1.1 Thrust-Shape Problem

Determine a shape of the cable, if the thrust of the system is given as $H = 24$ kN.

Vertical reactions of the cable:

$$R_A \rightarrow \sum M_B = 0 : \quad R_A = \frac{P(l-a)}{l} = \frac{20(25-10)}{25} = 12 \text{ kN},$$

$$R_B \rightarrow \sum M_A = 0 : \quad R_B = \frac{Pa}{l} = \frac{20 \cdot 10}{25} = 8 \text{ kN}$$

We can see that the vertical reactions do not depend on the value of the thrust H . It happens because supports A and B are located on the same elevation. The forces acting on the segment at support A and corresponding force polygon are shown in Fig. 7.3b. For assumed x - y -coordinate system the angles α_0 and α_1 belong to the fourth and first quadrant, respectively, so a shape of the cable is defined as follows:

$$\tan \alpha_0 = -\frac{R_A}{H} = -\frac{12}{24} = -\frac{1}{2} \rightarrow \cos \alpha_0 = \frac{2}{\sqrt{5}},$$

$$\tan \alpha_1 = \frac{R_B}{H} = \frac{8}{24} = \frac{1}{3} \rightarrow \cos \alpha_1 = \frac{3}{\sqrt{10}}$$

The y -ordinate of the cable at the location of the load P is

$$y = a \tan \alpha_0 = -10 \cdot 0.5 = -5 \text{ m}.$$

The negative sign corresponds to the adopted x - y -coordinate system. The sag at the point C is $f = 5$ m.

Tensions in the left and right portions of the cable may be presented in terms of thrust as follows:

$$N_{A-1} = \frac{H}{\cos \alpha_0} = \frac{24 \times \sqrt{5}}{2} = 26.83 \text{ kN}, \quad N_{B-1} = \frac{H}{\cos \alpha_1} = \frac{24 \times \sqrt{10}}{3} = 25.30 \text{ kN}$$

Increasing of the thrust H leads to decreasing of the angle α_0 and α_1 ; as a result, the sag of the cable is decreasing and tension in both parts of the cable is increasing.

The tensions in the left and right portions of the cable can be determined using expression (7.6).

Reference beam shear forces are $Q_{A-1} = R_A = 12 \text{ kN}$, $Q_{1-B} = R_A - P = -R_B = -8 \text{ kN}$. Therefore

$$N_{A-1} = \sqrt{Q_{A-1}^2 + H^2} = \sqrt{12^2 + 24^2} = 26.83 \text{ kN};$$

$$N_{1-B} = \sqrt{Q_{1-B}^2 + H^2} = \sqrt{(-8)^2 + 24^2} = 25.30 \text{ kN}$$

Now let us find sag using the concept of the reference beam (Fig. 7.3c). The bending moment at point C equals $M_C = \frac{Pab}{l} = \frac{(20 \times 10 \times 15)}{25} = 120 \text{ kNm}$. According to (7.4) the sag f at point C is $f_C = \frac{M_C^0}{H} = \frac{120}{24} = 5 \text{ m}$. Obtained result presents the *distance* between chord AB and cable and it is not related to adopted x - y -coordinate system.

7.2.1.2 Length-Thrust Problem

Determine a thrust of the cable, if total length L of the cable is given.

Length of the cable shown in Fig. 7.3a equals

$$L = \frac{a}{\cos \alpha_0} + \frac{l-a}{\cos \alpha_1} = a\sqrt{1 + \tan^2 \alpha_0} + (l-a)\sqrt{1 + \tan^2 \alpha_1}$$

According to force triangle (Fig. 7.3b), $\tan \alpha_0 = R_A/H = (P/H) \cdot (l-a)/l$; therefore length of the cable in terms of active force P and thrust H may be presented as follows:

$$L = a\sqrt{1 + \frac{P^2}{H^2} \left(\frac{l-a}{l}\right)^2} + (l-a)\sqrt{1 + \frac{P^2}{H^2} \frac{a^2}{l^2}}$$

Solving this equation with respect to H leads to the following expression for thrust in terms of P , L , l , and a :

$$H = P \frac{2l_0\delta(1-\delta)}{\sqrt{l_0^4 - 2l_0^2(2\delta^2 - 2\delta + 1) + 4\delta^2 - 4\delta + 1}}, \quad l_0 = \frac{L}{l}, \quad \delta = \frac{a}{l}$$

Let $\delta = 0.4$ and $L = 1.2l = 30 \text{ m}$, so $l_0 = 1.2$. In this case the thrust equals

$$H = P \frac{2 \times 1.2 \times 0.4(1-0.4)}{\sqrt{1.2^4 - 2 \times 1.2^2(2 \times 0.4^2 - 2 \times 0.4 + 1) + 4 \times 0.4^2 - 4 \times 0.4 + 1}} = 0.7339P$$

After that, the shape of the cable and internal forces in the cable may be defined easily.

The special case $a = 0.5l$ leads to the following results for thrust and sag:

$$H = \frac{P}{2\sqrt{l_0^2 - 1}},$$

$$f = \frac{Pl}{4H} = \frac{l}{2}\sqrt{l_0^2 - 1}$$

There are some interesting numerical results. Assume that $l_0 = \frac{L}{l} = 1.01$. In this case $f = 0.071l$, so if the total length of the cable L exceeds the span l only on 1%, then sag of the cable comprises 7% of the span.

7.2.2 Cable Subjected to Uniformly Distributed Load

Distributed load within the horizontal projection of the cable may be considered for case when the hangers (Fig. 7.1) are located very often. Design diagram of flexible cable under uniformly distributed load is presented in Fig. 7.4.

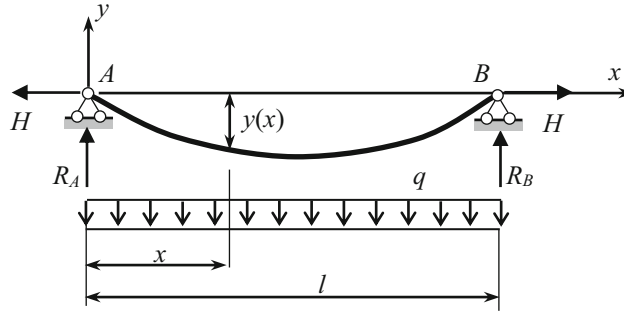


Fig. 7.4 A cable under uniformly distributed load

7.2.2.1 Thrust-Shape Problem

Thrust H of the cable is given. Determine the shape of the cable and calculate the distribution of internal forces. For solving of this problem we will use two approaches:

1. Integration of fundamental equation of a flexible cable (7.8) leads to the following expressions for slope and shape of the cable:

$$\begin{aligned}\frac{dy}{dx} &= \frac{q}{H}x + C_1 \\ y &= \frac{q}{H}\frac{x^2}{2} + C_1x + C_2\end{aligned}$$

Constant of integration is obtained from boundary conditions: at $x = 0$ (support A) $y = 0$ and at $x = l$ (support B) $y = 0$. These conditions applied to equation $y(x)$ lead to the following constants of integration: $C_2 = 0$ and $C_1 = -ql/2H$. Now the shape of a cable and slope in terms of load q and thrust H are described by equations

$$y(x) = -\frac{ql^2}{2H}\left(\frac{x}{l} - \frac{x^2}{l^2}\right) \quad (7.9a)$$

$$\tan \theta = \frac{dy}{dx} = \frac{ql}{2H}\left(2\frac{x}{l} - 1\right) \quad (7.9b)$$

Equation $y(x)$ presents symmetrical parabola. At $x = l/2$, maximum y -coordinate equals to

$$y_{\max} = -\frac{ql^2}{8H} \quad (7.10)$$

The tension N at any section x in terms of thrust H is

$$N(x) = \frac{H}{\cos \theta} = H\sqrt{1 + \tan^2 \theta} = H\sqrt{1 + \frac{q^2 l^2}{4H^2}\left(2\frac{x}{l} - 1\right)^2} \quad (7.11)$$

This equation may be obtained from (7.6), where shear $Q = R_A - qx$.

At the lowest point ($x = l/2$) a tension $N = H$. The maximum tension occurs at supports

$$N_{\max} = H\sqrt{1 + \frac{q^2 l^2}{4H^2}} \quad (7.12)$$

2. The concept of the reference beam leads to the following procedure. The bending moment for reference beam at any section x is $M_x^0 = (ql/2)x - (qx^2/2)$. According to (7.4) and taking into account the direction of the y -axis (Fig. 7.4) we immediately get the expression (7.9a) for $y(x)$, and as result, the formulas (7.9b, 7.10, 7.11, 7.12) for slope, $y_{\max}$, tension $N(x)$, and maximum tension $N_{\max}$.

7.2.2.2 Length-Thrust Problem

Expression for total length L of the cable is

$$L = \int_0^l \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad (7.13)$$

Since the sag of the cable is $f = ql^2/8H$, expression (7.9b) for slope at any x in terms of sag f may be presented as

$$\frac{dy}{dx} = \frac{ql}{2H} \left(\frac{2x}{l} - 1\right) = \frac{4f}{l} \left(\frac{2x}{l} - 1\right) \quad (7.14)$$

Therefore, the total length L of the cable in terms of sag according to (7.13) becomes

$$L = \int_0^l \sqrt{1 + \left(\frac{4f}{l}\right)^2 \left(\frac{2x}{l} - 1\right)^2} dx \quad (7.13a)$$

Approximate Solution of Length Determination

It is known that the radical can be represented in a series as

$$\sqrt{1 + \varepsilon} \cong 1 + \frac{1}{2}\varepsilon - \frac{1}{8}\varepsilon^2 + \dots \quad (\varepsilon \ll 1)$$

Therefore, the expression (7.13a) can be calculated as

$$L = \int_0^l \left\{ 1 + \frac{1}{2} \left[\frac{4f}{l} \left(\frac{2x}{l} - 1\right) \right]^2 - \frac{1}{8} \left[\frac{4f}{l} \left(\frac{2x}{l} - 1\right) \right]^4 + \dots \right\} dx \quad (7.15)$$

Integration of this relation leads to the following approximate expression:

$$L = l \left[1 + \frac{8}{3} \left(\frac{f}{l}\right)^2 - \frac{32}{5} \left(\frac{f}{l}\right)^4 + \frac{256}{7} \left(\frac{f}{l}\right)^6 - \dots \right] \quad (7.16)$$

This expression allows calculating the sag f of the cable in terms of total length L and span l . Since $f = ql^2/8H$, the length of the cable in terms of thrust H may be presented as follows:

$$L = l \left[1 + \frac{1}{24} \frac{q^2 l^2}{H^2} - \frac{1}{640} \frac{q^4 l^4}{H^4} + \dots \right] \quad (7.16a)$$

This expression allows calculating thrust H in terms of span l , total length of the cable L , and load q .

In case of gentle cable (if sag $f < 0.1l$) then it is sufficient to take into account only two terms in equation (7.16a):

$$L = l \left(1 + \frac{1}{24} \frac{q^2 l^2}{H^2} \right) \quad (7.16b)$$

Therefore, thrust in terms of total length of the cable L and span l may be presented in the form

$$H = \frac{ql}{2\sqrt{6}} \cdot \frac{1}{\sqrt{l_0 - 1}}, \quad l_0 = \frac{L}{l} \quad (7.16c)$$

The maximum y-coordinate of the cable in terms of total length L and span l is

$$y_{\max} = -\frac{\sqrt{3}}{2\sqrt{2}}l \cdot \sqrt{\frac{L}{l} - 1} \quad (7.16d)$$

Exact Solution of the Length Determination

Integrating of Eq. (7.13a) leads to the following exact expression for total length in terms of q , l and H :

$$L = \frac{l}{2} \sqrt{1 + \left(\frac{ql}{2H}\right)^2} + \frac{H}{q} \sinh^{-1} \frac{ql}{2H} \quad (7.16e)$$

Table 7.1 contains results of numerical solution of approximate equation (7.16a) for different total length L of the cable. Three and two terms of this equation have been held. Results of numerical solution are compared with the solution of exact Eq. (7.16e). The problems are solved for $q = 2$ kN/m and $l = 30$ m.

Table 7.1 Cable under uniformly distributed load. Thrust H kN of the cable vs. the total length L

	Total length of a cable L , m		
	32	34	36
Two terms	47.4342	33.5410	27.3861
Three terms	45.8884	31.1147	23.9464
Exact solution	46.0987	31.7556	25.3087

We can see that even two terms of Eq. (7.16a) lead to quite sufficient accuracy. Moreover, since $H_{3\text{terms}} < H_{\text{ex.sol}} < H_{2\text{terms}}$, two-term approximation is more preferable for design.

Let us consider the analysis of flexible cable using two approaches: (a) integrating of differential equation (7.8) and (b) the concept of reference beam.

Example 7.1 Design diagram of flexible cable with support points A and B on different levels is presented in Fig. 7.5. The cable is subjected to uniformly distributed load q along the horizontal line. Find the shape of the cable and determine the distribution of internal forces, if thrust H of the cable is given. Parameters of the system are $l = 30$ m, $c = 3$ m, $H = 40$ kN, and $q = 1.8$ kN/m.

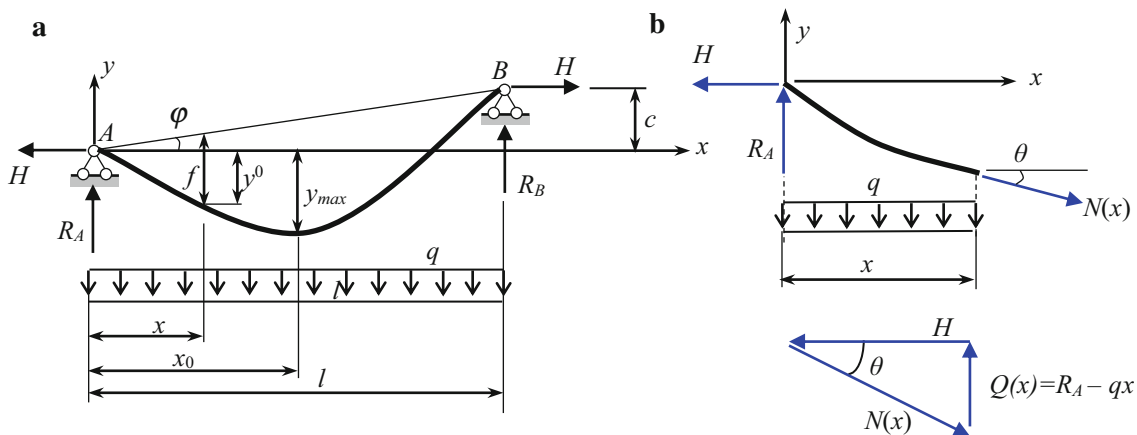


Fig. 7.5 (a) Cable under the action of uniformly distributed load; (b) free-body diagram of the part of the cable

We note the practical significance of this problem for the construction of a suspension bridge. Bearing structures—pylons of a suspension bridge—may have different heights due to different reasons (architectural requirements, landscape peculiarities, etc.). It is assumed that the permissible load (thrust) applied at the top of the pylon is known. The bridge deck (it may be

slab with concave surface) is attached to a cable using a series of vertical hangers. The problem is to determine the lengths of all hangers that correspond to the calculated shape of the cable.

Solution (a) Differential equation of a flexible cable is $\frac{d^2y}{dx^2} = \frac{q}{H}$; its integration leads to the following expressions for slope and shape:

$$\begin{aligned}\frac{dy}{dx} &= \frac{q}{H}x + C_1 \\ y &= \frac{q}{H}\frac{x^2}{2} + C_1x + C_2\end{aligned}$$

Constants of integration should be calculated from the following boundary conditions: at $x = 0$ (support A) $y = 0$ and at $x = l$ (support B) $y = c$. Constants of integration are $C_2 = 0$ and $C_1 = \left(c - \frac{ql^2}{2H}\right) \cdot \frac{1}{l}$. Now the shape of the cable and its slope for any x may be presented in the form

$$\begin{aligned}y(x) &= \frac{ql^2}{2H} \left(\frac{x^2}{l^2} - \frac{x}{l} \right) + c \frac{x}{l} = \frac{1.8 \times 30^2}{2 \times 40} \left(\frac{x^2}{30^2} - \frac{x}{30} \right) + 3 \frac{x}{30} = 0.675 \left(\frac{x^2}{30} - x \right) + 0.1x \\ \tan \theta &= \frac{dy}{dx} = \frac{ql}{2H} \left(2 \frac{x}{l} - 1 \right) + \frac{c}{l} = \frac{1.8 \times 30}{2 \times 40} \left(2 \frac{x}{30} - 1 \right) + \frac{3}{30} = 0.675 \left(\frac{x}{15} - 1 \right) + 0.1\end{aligned}$$

Equation $y(x)$ presents nonsymmetrical parabola. In the lowest point the slope of the cable equals zero:

$$\tan \theta = \frac{ql}{2H} \left(2 \frac{x}{l} - 1 \right) + \frac{c}{l} = 0,$$

This equation leads to corresponding x_0 -ordinate:

$$x_0 = \frac{H}{q} \left(\frac{ql}{2H} - \frac{c}{l} \right) = \frac{40}{1.8} \left(\frac{1.8 \times 30}{2 \times 40} - \frac{3}{30} \right) = 12.78 \text{ m}$$

The maximum y occurs at x_0 :

$$y_{\max} = y(12.78) = 0.675 \left(\frac{12.78^2}{30} - 12.78 \right) + 0.1 \times 12.78 = -3.674 \text{ m}$$

Reaction of support A equals

$$R_A \rightarrow \sum M_B = 0 : -R_A l - Hc + \frac{ql^2}{2} = 0 \rightarrow R_A = \frac{ql}{2} - \frac{Hc}{l}$$

Free-body diagram of the left part of the cable and corresponding force triangle are presented in Fig. 7.5b. Since shear $Q(x) = R_A - qx$, a tension at any section according to (7.6) equals

$$N(x) = \sqrt{H^2 + (R_A - qx)^2} = \sqrt{H^2 + \left(\frac{ql}{2} - \frac{Hc}{l} - qx \right)^2}$$

The tension in the lowest point:

$$N(x_0) = \sqrt{H^2 + \left(\frac{ql}{2} - \frac{Hc}{l} - qx_0 \right)^2} = \sqrt{H^2 + \left(\frac{1.8 \times 30}{2} - \frac{40 \times 3}{30} - 1.8 \times 12.78 \right)^2} = \sqrt{H^2 + (27 - 27)^2} = H$$

Maximum tension occurs at supports

$$N(0) = \sqrt{40^2 + \left(\frac{1.8 \times 30}{2} - \frac{40 \times 3}{30} - 1.8 \times 0 \right)^2} = 46.14 \text{ kN}$$

$$N(l) = \sqrt{40^2 + \left(\frac{1.8 \times 30}{2} - \frac{40 \times 3}{30} - 1.8 \times 30 \right)^2} = 50.60 \text{ kN}$$

(b) The reaction of the reference beam is $R_A^0 = \frac{ql}{2}$. The bending moment of the reference beam and function $f(x)$ measured from the inclined chord AB are

$$M^0(x) = \frac{ql}{2}x - \frac{qx^2}{2}$$

$$f(x) = \frac{M^0(x)}{H} = \frac{1}{H} \left(\frac{ql}{2}x - \frac{qx^2}{2} \right)$$

Distance y^0 between the horizontal line x and cable becomes

$$y^0(x) = f(x) - x \cdot \tan \varphi = \frac{1}{H} \left(\frac{ql}{2}x - \frac{qx^2}{2} \right) - \frac{c}{l}x$$

Condition $\frac{dy^0}{dx} = 0$ leads to the parameter $x_0 = 12.78$ m obtained above. The distance between the cable in the lowest point and horizontal line becomes

$$y^0(x_0) = \frac{1}{40} \left(\frac{1.8 \times 30}{2} 12.78 - \frac{1.8 \times 12.78^2}{2} \right) - \frac{3}{30} \times 12.78 = 3.674 \text{ m}$$

This result had been obtained above.

7.3 Effect of Arbitrary Load on the Thrust and Sag

So far we considered the behavior of cables which are subjected to concentrated or distributed load only. In case of different loads acting simultaneously, formula (7.4) is most appropriate. However in engineering practice another type of loading is possible, mainly: the cable is subjected to any dead load and after that additional live load is applied. How will this change the shape and state of the cable? This problem may be effectively solved by knowing the expression related to the total length of the cable, thrust, and external loads.

Consider a cable supported at points A and B and subjected to any loads (Fig. 7.6). Assume that a cable is inextensible, and support points of the cable do not have the mutual displacements. The covered span is l , while a total unstressed length of a cable is L .

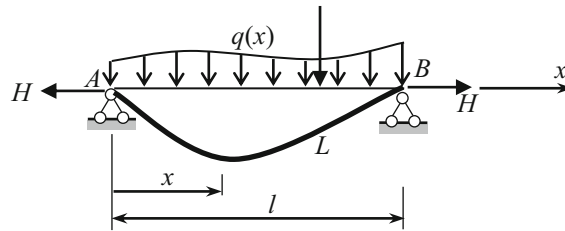


Fig. 7.6 Cable carrying arbitrary dead and live load

The total length of the cable is defined by exact Eq. (7.13). For gentle cable a slope is small, so

$$\sqrt{1 + \left(\frac{dy}{dx} \right)^2} \cong 1 + \frac{1}{2} \left(\frac{dy}{dx} \right)^2 \quad (7.17)$$

and expression for L becomes

$$L = \int_0^l \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \cong \int_0^l \left[1 + \frac{1}{2} \left(\frac{dy}{dx}\right)^2\right] dx \quad (7.18)$$

According to (7.4), the shape of the cable is defined by the expression $y = M/H$, where M is a bending moment of the reference beam, and H is a thrust. Therefore the slope can be calculated as follows:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{M}{H}\right) = \frac{1}{H} \frac{dM}{dx} = \frac{Q(x)}{H}$$

where $Q(x)$ is shear at any section of the reference beam. After that expression (7.18) becomes

$$L = \int_0^l \left[1 + \frac{1}{2} \left(\frac{Q(x)}{H}\right)^2\right] dx = l + \frac{1}{2H^2} \int_0^l Q^2(x) dx$$

Usually this formula is written in the following form:

$$L = l + \frac{D}{2H^2}, \quad D = \int_0^l Q^2(x) dx, \quad (7.19)$$

Table A.19 contains the load characteristic D for most important cases of loading.

Equation (7.19) allows presenting the thrust H in terms of load characteristic D as follows:

$$H = \sqrt{\frac{D}{2l(l_0 - 1)}}, \quad l_0 = \frac{L}{l} \quad (7.20)$$

It is possible to show that if support points A and B of the cable are located on the different elevation then

$$L = \frac{l}{\cos \varphi} + \frac{D}{2H^2} \cos^3 \varphi$$

where φ is angle between chord AB and x -axis.

Formulas (7.18)–(7.20) entirely solve the problem of determination of the thrust of inextensible gentle cable, subjected to arbitrary vertical load. These formulas are approximate since for their deriving the approximate relationship (7.17) has been used.

Application of expression (7.20) for some classical loading cases is shown below.

1. A cable with total length L and span l carries a concentrated force P throughout distance a starting from the left end of the cable (Fig. 7.3). Let dimensionless parameters be $l_0 = \frac{L}{l} = 1.2$ and $\xi = \frac{a}{l} = 0.4$. Load characteristic D (Table A.19, row 1) is $D = P^2 l \xi (1 - \xi) = P^2 l \times 0.4(1 - 0.4) = 0.24 P^2 l$. The thrust H according to (7.20) equals

$$H = \sqrt{\frac{0.24 P^2 l}{2l(1.2 - 1)}} = 0.7746 P$$

The exact solution is $H = 0.7339 P$ (Sect. 7.2). Error is 5.54%.

2. The cable of total length L and span l carries a uniformly distributed load q (Fig. 7.4). Load characteristic is $D = \frac{q^2 l^3}{12}$ (Table A.19). According to formula (7.16)

$$l_0 = \frac{L}{l} \cong 1 + \frac{8}{3} \frac{f^2}{l^2}$$

Therefore the thrust becomes

$$H = \sqrt{\frac{D}{2l(l_0 - 1)}} = \sqrt{\frac{q^2 l^3}{12 \times 2l \left(1 + \frac{8}{3} \frac{f^2}{l^2} - 1\right)}} = \frac{ql^2}{8f}$$

This exact result has been obtained earlier by using formula (7.10).

The formulas above allow considering very important problem for determining how the sag and thrust of the cable will *change* if additional live loads are applied.

Example 7.2 The flexible, inextensible, and gentle cable is suspended between two absolutely rigid supports of same elevation. The total length of the cable is L and span equals l . The cable is subjected to uniformly distributed dead load q within the horizontal line; corresponding thrust and sag are H_q and f_q . Calculate a *change* of the thrust and sag of the cable, if the entire cable upon the load q is loaded by additional concentrated live load P at the axis of symmetry of the cable (Fig. 7.7).

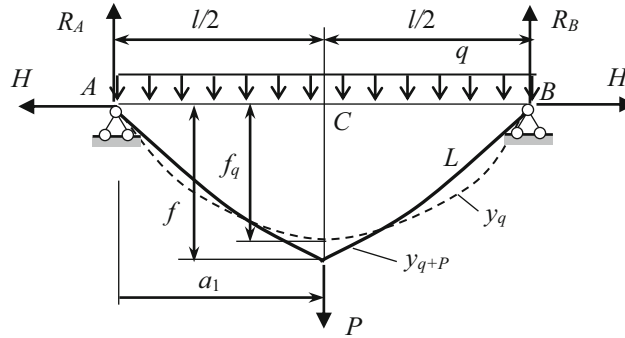


Fig. 7.7 Cable carrying uniformly distributed load q with additional force P

Solution If a cable is loaded by uniformly distributed load q only then load characteristic is $D_q = \frac{q^2 l^3}{12}$.

Corresponding thrust and sag are H_q and f_q . In this case Eq. (7.19) becomes

$$L = l + \frac{1}{2H_q^2} \cdot \frac{q^2 l^3}{12} \quad (a)$$

If additional load P is applied at the middle of the cable then load characteristic accordingly (Table A.19, line 5) becomes

$$D_{q+P} = \frac{q^2 l^3}{12} (1 + 3\gamma + 3\gamma^2), \quad \gamma = \frac{P}{ql}$$

Thus (7.19) becomes

$$L = l + \frac{1}{2H_{q+P}^2} \cdot \frac{q^2 l^3}{12} (1 + 3\gamma + 3\gamma^2), \quad (b)$$

where H_{q+P} is thrust due to both loads q and P .

Since the cable is inextensible left parts of the (a) and (b) are same; therefore

$$l + \frac{1}{2H_q^2} \cdot \frac{q^2 l^3}{12} = l + \frac{1}{2H_{q+P}^2} \cdot \frac{q^2 l^3}{12} (1 + 3\gamma + 3\gamma^2)$$

which leads to the required relationship

$$H_{q+P} = H_q \sqrt{1 + 3\gamma + 3\gamma^2}$$

The sag of the cable

$$f_{q+P} = \frac{M_C^{\text{beam}}}{H_{q+P}} = \frac{\frac{ql^2}{8} + \frac{Pl}{4}}{H_q \sqrt{1 + 3\gamma + 3\gamma^2}} = f_q \frac{1 + 2\gamma}{\sqrt{1 + 3\gamma + 3\gamma^2}}, \quad f_q = \frac{ql^2}{8H_q}$$

Let $\gamma = Pl/ql = 0.2$. In this case the thrust and sag of the cable are

$$H_{q+P} = H_q \sqrt{1 + 3 \times 0.2 + 3 \times 0.2^2} = 1.31H_q$$

$$f_{q+P} = f_q \frac{1 + 2 \times 0.2}{\sqrt{1 + 3 \times 0.2 + 3 \times 0.2^2}} = 1.068f_q$$

Thus application of additional concentrated force $P = 0.2 ql$ leads to increasing of the thrust and sag of the non-extensible cable on 31% and 6.8%, respectively.

7.4 Cable with Self-Weight

This section is devoted to the analysis of cable carrying a load uniformly distributed along the cable itself. Relationships between parameters of cable, its length, shape, and internal forces are developed. The direct and inverse problems are considered.

7.4.1 Fundamental Relationships

A cable is supported at points A and B and loaded by uniformly distributed load q_0 along the cable *itself*. This load presents not only dead load (self-weight of the cable), but also a live load, such as a glazed ice. Cables of this type are called the *catenary* (Latin word *catena* means a “chain”). It was named by Huygens in 1691.

Design diagram of a cable is presented in Fig. 7.8. The y -axis passes through the lowest point C . The following notations are used: s is curvilinear coordinate along the cable; $W = q_0 s$ is a total weight of portion s ; N is a tension of the cable and H presents a thrust of the cable.

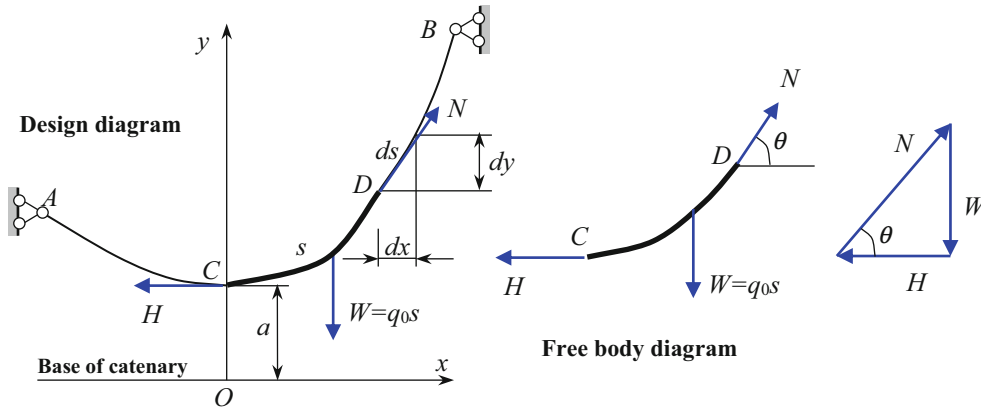


Fig. 7.8 Design diagram of the catenary. Free-body diagram for portion CD and force triangle

For any point of the cable, relationship between N , H , and W obeys the law of force triangle:

$$N = \sqrt{H^2 + W^2} \quad (7.21)$$

Expression (7.21) may be conveniently expressed in terms of additional parameter $a = \frac{H}{q_0}$. This parameter is measured from the lowest point C of the cable and defines an origin O ; the line $O-x$ is called a base of catenary. In terms of a and s the normal force becomes

$$N = q_0 \sqrt{a^2 + s^2} \quad (7.22)$$

The relationship between coordinates x and s in differential form is

$$dx = ds \cos \theta = ds \cdot \frac{H}{N} = \frac{aq_0}{q_0 \sqrt{a^2 + s^2}} ds = \frac{ads}{\sqrt{a^2 + s^2}}$$

Integrating this equation from point $C(0, a)$ to $D(x, y)$ allows calculating the x -coordinate of any point of the cable in terms of s and $a = H/q_0$:

$$x = \int_0^s \frac{ads}{\sqrt{a^2 + s^2}} = a \sinh^{-1} \frac{s}{a} \quad (7.23)$$

The inverse relation, i.e., the solution of (7.23) with respect to s , is

$$s = a \sinh \frac{x}{a} \quad (7.24)$$

This formula connects the curvilinear coordinate s with x -coordinate and parameter a , and allows obtaining some useful relationships in terms of intensity load q_0 and thrust H .

The length of the cable from lowest point C to $D(x, y)$ is

$$L_{C-D} = \frac{H}{q_0} \sinh \left(\frac{q_0}{H} x \right) \quad (7.25)$$

Formula (7.22) for tension at any point can be rewritten as follows:

$$N(x) = H \sqrt{1 + \sinh^2 \frac{q_0}{H} x} = H \cosh \frac{q_0}{H} x \quad (7.26)$$

The minimum tension $N = H$ occurs at $x = 0$, to say in the lowest point.

In order to obtain the equation of the catenary, we need to express y in terms of x :

$$dy = dx \cdot \tan \theta = \frac{W}{H} dx = \frac{q_0 s}{aq_0} dx = \sinh \frac{x}{a} dx$$

After integrating this equation from point C to point D , we get

$$\int_a^y dy = \int_0^x \sinh \frac{x}{a} dx \rightarrow y|_a^y = a \cosh \frac{x}{a} \Big|_0^x$$

The equation of the curve, assumed by the cable, and corresponding slope at any point are

$$\begin{aligned} y(x) &= a \cosh \frac{x}{a} = \frac{H}{q_0} \cosh \frac{q_0}{H} x \\ \tan \theta &= \frac{dy}{dx} = \sinh \frac{q_0}{H} x \end{aligned} \quad (7.27)$$

The express for slope may be immediately obtained from a force triangle (Fig. 7.8):

$$\tan \theta = \frac{W}{H} = \frac{q_0 s}{H} = \frac{q_0}{H} \cdot a \sinh \frac{q_0}{H} x = \sinh \frac{q_0}{H} x$$

Below is presented the analysis of typical cases of cables carrying a uniformly distributed load along the cable itself:

1. Supports are located at the same level.
2. Supports are located at different elevations and saddle point is within the span.
3. Supports are located on different elevations and cable does not have a saddle point within the span.

7.4.2 Cable with Supports Located at the Same Level

In this case a cable has the axis of symmetry. It is pertinent here to derive the simple and useful expression for maximum tension for the cable in terms of q_0 , f , and H . Maximum tension $N_{\max}$ occurs at the supports ($x = \pm l/2$); then

$$N_{\max} = H \sqrt{1 + \sinh^2 \frac{q_0 l}{2H}} \rightarrow \sinh^2 \frac{q_0 l}{2H} = \frac{N_{\max}^2}{H^2} - 1 \quad (7.28)$$

The sag f of the cable at the axis of symmetry ($x = 0$) in terms of ordinate y at support B (i.e., at $x = l/2$) is

$$f = y\left(\frac{l}{2}\right) - a = \frac{H}{q_0} \cosh \frac{q_0 l}{2H} - \frac{H}{q_0} = \frac{H}{q_0} \left(\cosh \frac{q_0 l}{2H} - 1 \right) \rightarrow \cosh^2 \frac{q_0 l}{2H} = \left(\frac{q_0 f}{H} + 1 \right)^2 \quad (7.29)$$

Since $\sinh^2 z = \cosh^2 z - 1$, expressions (7.28) and (7.29) lead to the following simple and helpful formula:

$$N_{\max} = q_0 f + H \quad (7.30)$$

Example 7.3 A uniform cable weighting $q_0 = 1.25$ kN/m is suspended between two points A and B of equal elevation, which are $l = 20$ m apart as shown in Fig. 7.9.

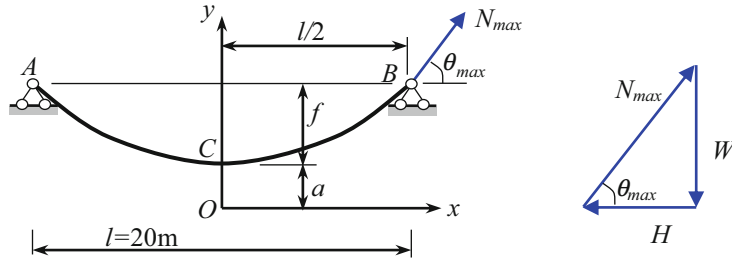


Fig. 7.9 Cable carrying a uniformly distributed load along the cable itself. Design diagram and force triangle for CB portion

Thrust-shape problem. Determine the shape of the cable, distribution of tension, and length of the cable, if a thrust is $H = 5.6$ kN.

Solution Parameter of catenary $a = \frac{H}{q_0} = \frac{5.6}{1.25} = 4.48$ m. Equations of the curve and slope assumed by the cable are

$$\begin{aligned} y(x) &= \frac{H}{q_0} \cosh \frac{q_0}{H} x = 4.48 \cosh (0.2232x) \\ \tan \theta &= \sinh \frac{q_0}{H} x = \sinh (0.2232x) \end{aligned} \quad (a)$$

Coordinates of point B are $x = 10$ m, $y(10) = 4.48 \cosh (0.2232 \times 10) = 21.11$ m. Therefore, sag of the cable at point C is $f = y(10) - a = 21.11 - 4.48 = 16.63$ m. Slope at point B is

$$\tan \theta_{\max} = \sinh(0.2232 \times 10) = 4.6056 \rightarrow$$

$$\sin \theta_{\max} = \sin(\tan^{-1} 4.6056) = 0.9772, \quad \cos \theta_{\max} = \cos(\tan^{-1} 4.6056) = 0.2122$$

Tension at any point is $N = H \sqrt{1 + \sinh^2 \frac{q_0}{H} x}$, and maximum tension occurs at point B and equals

$$N_{\max} = 5.6 \sqrt{1 + \sinh^2(0.2232 \times 10)} = 26.39 \text{ kN}$$

Since shape of the cable is symmetrical, the total length of the cable according to (7.25) is

$$L = 2a \sinh \frac{x}{a} \Big|_{x=\frac{l}{2}} = 2 \times 4.48 \sinh \frac{10}{4.48} = 41.27 \text{ m}$$

Control: If total length $L = 41.27$ m, then total weight of the cable equals $41.27 \times 1.25 = 51.58$ kN, and force W , considering portion CB , equals $51.58/2 = 25.79$ kN. From a force triangle we can calculate

$$H = N_{\max} \cos \theta_{\max} = 26.39 \times 0.2122 = 5.6 \text{ kN}$$

$$W = N_{\max} \sin \theta_{\max} = 26.39 \times 0.9772 = 25.79 \text{ kN}$$

$$\tan \theta_{\max} = \frac{W}{H} = \frac{25.79}{5.6} = 4.605$$

The maximum tension according to (7.30) equals

$$N_{\max} = q_0 f + H = 1.25 \times (21.11 - 4.48) + 5.6 = 26.39 \text{ kN}$$

Length-thrust problem. Determine the thrust and shape of the cable, if a total length $L = 42$ m.

Solution We assume that a total length L of a cable is given and it is required to find the thrust, shape of the cable, and sag. Since the cable is symmetrical with respect to y -axis, the total length of the cable equals $L = 2a \sinh(x_B/a)$. Assume that the length of the cable for this problem is $L = 42$ m and $x_B = l/2 = 10$ m. Unknown parameter a may be calculated from transcendental equation (7.25), i.e.,

$$42 = 2a \sinh \frac{10}{a}$$

Solution of this equation leads to $a = 4.42$ m. Thrust of the cable is $H = a q_0 = 4.42 \times 1.25 = 5.525$ kN.

Equation of the curve and ordinate y for support points are

$$y(x) = \frac{H}{q_0} \cosh \frac{q_0}{H} x = 4.42 \cosh 0.22642x$$

$$y_{\max} = 4.42 \cosh \frac{1.25}{5.525} \times 10 = 21.46 \text{ m}$$

Sag of the cable at lowest point C is $f = 21.46 - 4.42 = 17.04$ m.

7.4.3 Cable with Supports Located on Different Elevations

In this case the shape of the cable is nonsymmetrical and position of the lowest point C is not known ahead. Availability of only two parameters, such as load per unit length of the cable q_0 and span l , is not enough for cable analysis. For determination of location of point C we need to formulate additional condition for curve $y(x)$ passing through points A and B . One of the additional parameters may be either value of thrust H or total length L or coordinates of any additional point of the cable.

Two different cases for position of saddle point are considered, i.e., saddle point within or outside of the span.

7.4.3.1 Saddle Point Within the Span

The following example illustrates the thrust-length procedure for analysis of the cable and corresponding peculiarities.

Example 7.4 A uniform cable with self-weight $q_0 = 1.25 \text{ kN/m}$ is suspended between two points $l = 20 \text{ m}$ apart horizontally, with one point of support 18 m higher than the other as shown in Fig. 7.10. Thrust of the cable is $H = 5.6 \text{ kN}$. Determine the shape of the cable, tension at supports, and total length of the cable. Assume that a saddle point is placed between supports.

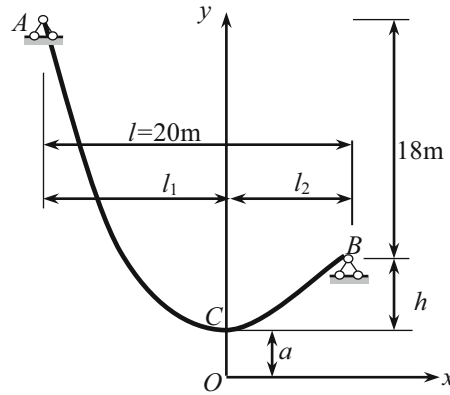


Fig. 7.10 Cable, carrying a uniformly distributed load along the cable itself

Solution Let a vertical y -axis pass through the lowest point C . Location of this point is defined by parameters l_1 and l_2 which are unknown in advance, while $l_1 + l_2 = 20 \text{ m}$. For this direct problem the parameter of catenary (Fig. 7.10) may be calculated right away, i.e., $a = H/q_0 = 5.6/1.25 = 4.48 \text{ m}$.

Shape of the cable is defined by the equation

$$y(x) = \frac{H}{q_0} \cosh \frac{q_0}{H} x = 4.48 \cosh (0.2232x) \quad (\text{a})$$

Conditions of the catenary passing through two points A and B are as follows:

$$\text{At } x = l_2 (\text{point } B) \text{ the ordinate } y = a + h = 4.48 + h$$

$$\text{At } x = -l_1 = -(20 - l_2) (\text{point } A) \text{ the ordinate } y = a + h + 18$$

Therefore, (a) for points B and A becomes

$$\text{Point } B : 4.48 \cosh (0.2232l_2) = 4.48 + h \quad (\text{b})$$

$$\text{Point } A : 4.48 \cosh [-0.2232(20 - l_2)] = 4.48 + h + 18$$

Equations (b) contain two unknowns h and l_2 . Expressing from both equations unknown parameter h and after that equating them, we obtain

$$4.48 \cosh (0.2232l_2) = 4.48 \cosh (0.2232l_2 - 4.464) - 18$$

The root of this equation is $l_2 = 8.103 \text{ m}$, which leads to $l_1 = l - l_2 = 11.897 \text{ m}$. Therefore, vertical distance h between points C and B according to formula (b) equals

$$h = 4.48 \cosh (0.2232 \times 8.103) - 4.48 = 9.555 \text{ m}$$

Now we can consider the right and left parts of the cable separately.

Curve CB. Equations of the curve and slope are

$$y(x) = 4.48 \cosh (0.2232x),$$

$$\tan \theta = \frac{dy}{dx} = \sinh (0.2232x)$$

Ordinate y and slope at the point B ($x_B = l_2 = 8.103 \text{ m}$) are

$$\begin{aligned}
y_B &= 4.48 \cosh(0.2232 \times 8.103) = 14.034 \text{ m}, \\
\tan \theta_B &= \sinh(0.2232 \times 8.103) = 2.969, \\
\theta_B &= 71.38^\circ; \quad \sin \theta_B = 0.9476.
\end{aligned}$$

The tension at the point B is

$$N_B = H \sqrt{1 + \sinh^2 \frac{q_0}{H} x_B} = 5.6 \sqrt{1 + \sinh^2(0.2232 \times 8.103)} = 17.544 \text{ kN}$$

Length of the portion CB is

$$L_{CB} = a \sinh \frac{x_B}{a} = 4.48 \sinh \frac{8.103}{4.48} = 13.303 \text{ m}$$

Curve AC . Equations of the curve and slope are

$$\begin{aligned}
y(x) &= 4.48 \cosh(0.2232x), \\
\tan \theta &= \frac{dy}{dx} = \sinh(0.2232x)
\end{aligned}$$

Ordinate y and slope at the point A ($x_A = -l_1 = -11.897 \text{ m}$) are

$$\begin{aligned}
y_A &= 4.48 \cosh(0.2232 \times (-11.897)) = 32.034 \text{ m}, \\
\tan \theta_A &= \sinh(0.2232 \times (-11.897)) = -7.0796, \\
\theta_A &= 81.96^\circ; \quad \sin \theta_A = 0.990.
\end{aligned}$$

Calculated coordinates y for support points A and B satisfy the given design diagram:

$$y_A - y_B = 32.034 - 14.034 = 18 \text{ m}$$

Tension at point A is

$$N_A = H \sqrt{1 + \sinh^2 \frac{q_0}{H} x_A} = 5.6 \sqrt{1 + \sinh^2(0.2232 \times (-11.897))} = 40.04 \text{ kN}$$

Length of the portion CA and total length of the cable are

$$\begin{aligned}
L_{CA} &= a \sinh \frac{x_A}{a} = 4.48 \sinh \frac{11.897}{4.48} = 31.725 \text{ m}, \\
L &= L_{AC} + L_{CB} = 31.725 + 13.303 = 45.028 \text{ m}
\end{aligned}$$

Control: The tension at the support points and total weight of the cable must satisfy the condition

$$\sum Y = N_A \sin \theta_A + N_B \sin \theta_B - Lq_0 = 0$$

In our case we have

$$N_A \sin \theta_A + N_B \sin \theta_B - Lq_0 = 40.04 \times 0.990 + 17.544 \times 0.9476 - 45.028 \times 1.25 = 56.264 - 56.285$$

The relative error equals 0.035%.

7.4.3.2 Saddle Point Outside of the Span

Now let us consider the cable with total length L_0 , which is suspended between two points A and B , as shown in Fig. 7.11. Peculiarity of this design diagram, unlike the previous case 2, is that the curve AB of the cable has no point, for which

$\tan\theta = 0$, since the lowest point C is located beyond the curve AB . Let the distance between support points in horizontal and vertical directions be D_0 and h_0 , respectively; q_0 is weight per unit length. It is required to determine the shape of the cable, thrust H , and tension N_A and N_B at supports.

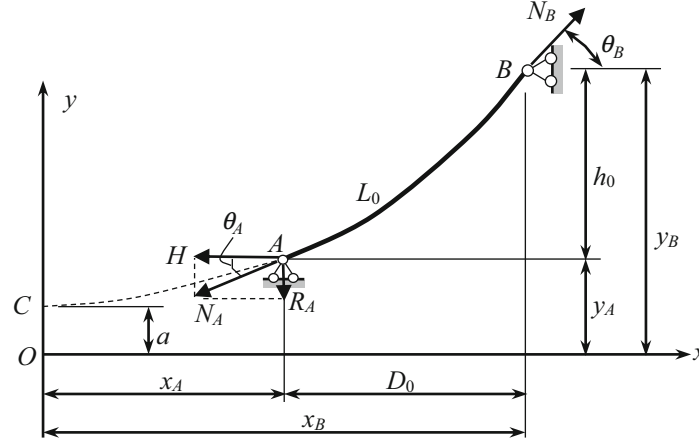


Fig. 7.11 Design diagram of the catenary; saddle point C beyond the span

For this length-thrust problem parameter $a = H/q_0$ cannot be calculated right now because thrust H is unknown yet. However, this parameter may be calculated analytically having the total length of the cable L_0 and dimensions D_0 and h_0 . For this some steps should be performed beforehand.

1. The length of the curve from point C to any point (x, y) according to (7.25) is $L_{C-x} = a \sinh\left(\frac{x}{a}\right)$, so the length of curves CB and CA is

$$L_{CB} = a \sinh \frac{x_B}{a}, \quad L_{CA} = a \sinh \frac{x_A}{a}$$

The total length AB of a cable is

$$L_0 = L_{CB} - L_{CA} = a \sinh \frac{x_B}{a} - a \sinh \frac{x_A}{a} \quad (7.31)$$

2. Equation of the curve assumed by cable according to (7.27) is $y(x) = a \cosh(x/a)$. Therefore, ordinates of points B and A are

$$y_B = a \cosh \frac{x_B}{a}, \quad y_A = a \cosh \frac{x_A}{a}$$

The vertical distance between two supports is

$$h_0 = y_B - y_A = a \cosh \frac{x_B}{a} - a \cosh \frac{x_A}{a} \quad (7.32)$$

Equations (7.31) and (7.32) present relationships between parameters $a = H/q_0$, L_0 , h_0 , and $D_0 = x_B - x_A$ and they contain two unknown parameters: a and x_B (or x_A).

For given geometry parameters L_0 , D_0 , and h_0 , the analytical solution of Eqs. (7.31) and (7.32) leads to the following results.

Parameter a is determined from transcendental equation

$$\cosh \frac{D_0}{a} = 1 + \frac{L_0^2 - h_0^2}{2a^2} \quad (7.33)$$

or in equivalent form

$$\sinh \frac{D_0}{2a} = \frac{1}{2a} \sqrt{L_0^2 - h_0^2} \quad (7.33a)$$

Coordinates x_A and x_B of the points A and B are

$$x_A = a \tanh^{-1} \left(\frac{h_0}{L_0} \right) - \frac{D_0}{2}, \quad x_B = x_A + D_0 \quad (7.34)$$

The thrust of the cable equals $H = q_0 a$. Also, equations of the curve and slope according to (7.27) are

$$y(x) = a \cosh \frac{x}{a}; \quad \tan \theta(x) = \sinh \frac{x}{a} \quad (7.35)$$

Slopes at support points A and B are

$$\tan \theta_A = \sinh \frac{x_A}{a}; \quad \tan \theta_B = \sinh \frac{x_B}{a} \quad (7.35a)$$

Tensions at points A and B according to (7.1), (7.2), and (7.34) are

$$N_A = H \sqrt{1 + \sinh^2 \frac{x_A}{a}}; \quad N_B = H \sqrt{1 + \sinh^2 \frac{x_B}{a}}. \quad (7.36)$$

Example 7.5 Let us consider the cable, which is presented in Fig. 7.11. The cable has the following parameters: $L_0 = 117.7$ m; $D_0 = 100$ m; $h_0 = 57.7$ m; $q_0 = 0.014$ kN/m. Determine the shape of the cable, thrust, and tension.

Solution Parameter a of the catenary can be calculated from transcendental equation (7.33a):

$$\sinh \frac{100}{2a} = \frac{1}{2a} \sqrt{117.7^2 - 57.7^2} \rightarrow a = 127.41 \text{ m}$$

Coordinate x of the support A according to (7.34) is

$$x_A = 127.41 \cdot \tanh^{-1} \frac{57.7}{117.7} - \frac{100}{2} = 18.34 \text{ m}$$

Slopes at supports A and B according to (7.35a) are

$$\begin{aligned} \tan \theta_A &= \sinh \frac{18.34}{127.41} = 0.1446; & \sin \theta_A &= 0.1431; & \cos \theta_A &= 0.9897 \\ \tan \theta_B &= \sinh \frac{18.34 + 100}{127.41} = 1.0694; & \sin \theta_B &= 0.7304; & \cos \theta_B &= 0.6830 \end{aligned}$$

Thrust of the cable is

$$H = q_0 a = 0.014 \times 127.41 = 1.7837 \text{ kN}.$$

Expression (7.36) allows calculating the tension at supports A and B :

$$\begin{aligned} N_A &= 1.7837 \sqrt{1 + \sinh^2 \frac{18.34}{127.41}} = 1.802 \text{ kN}; \\ N_B &= 1.7837 \sqrt{1 + \sinh^2 \frac{18.34 + 100}{127.41}} = 2.609 \text{ kN} \end{aligned}$$

Control: Ordinates of support points are

$$y_A = a \cosh \frac{x_A}{a} = 127.41 \cosh \frac{18.34}{127.41} = 128.73 \text{ m};$$

$$y_B = a \cosh \frac{x_B}{a} = 127.41 \cosh \frac{18.34 + 100}{127.41} = 186.43 \text{ m}$$

Vertical distance between points A and B equals $h_0 = y_B - y_A = 186.43 \text{ m} - 128.73 \text{ m} = 57.7 \text{ m}$. This value was known by the condition of a problem.

The tension at the support points and total weight of the cable must satisfy the condition

$$\sum Y = -N_A \sin \theta_A + N_B \sin \theta_B - L_0 q = 0$$

In our case, for cable in whole we have the following result:

$$\sum Y = -1.802 \times 0.1431 + 2.609 \times 0.7304 - 117.7 \times 0.014 = -1.9057 + 1.9056 \cong 0$$

Trust can be calculated by formulas

$$H = N_A \cos \theta_A = 1.802 \times 0.9897 = 1.783 \text{ kN},$$

$$H = N_B \cos \theta_B = 2.609 \times 0.6830 = 1.782 \text{ kN}$$

Peculiarity of the problem with saddle point outside the cable is that the vertical reaction of the support A is directed *downward* as shown in Fig. 7.11.

7.5 Comparison of Parabolic and Catenary Cables

Let us compare the main results for parabolic and catenary cables. Assume that both cables are inextensible, and supported on same elevation. Both cables have the same span l , subjected to uniformly distributed load q . If the load q is distributed within the horizontal, then a curve of the cable is parabola (parabolic shape); if the load is distributed within the cable itself, then the curve of the cable is catenary. Table 7.2 contains some fundamental parameters for both cables. They are dimensionless sag-span ratio f/l , slope $\tan \theta$ at the supports, and dimensionless maximum cable tension-thrust ratio $N_{\max}/H$. Correspondence formulae are presented in brackets. The formulas for parabolic cable may be derived if hyperbolic functions are presented as series $\sinh z = z + \left(\frac{z^3}{3!}\right) + \dots$; $\cosh z = 1 + \left(\frac{z^2}{2!}\right) + \dots$.

Table 7.2 Comparison of parabolic and catenary cables

	Parabolic cable	Catenary
$\tan \theta_{\max}$	$\frac{ql}{8H}$ (7.10)	$\frac{H}{ql} \left(\cosh \frac{ql}{2H} - 1 \right)$ (7.29)
$\tan \theta_{\max}$	$\frac{ql}{2H}$ (7.9b)	$\sinh \frac{ql}{2H}$ (7.27)
$\frac{N_{\max}}{H}$	$\sqrt{1 + \left(\frac{ql}{2H}\right)^2}$ (7.12)	$\sqrt{1 + \sinh^2 \frac{ql}{2H}}$ (7.28)

Some numerical results are presented in Figs. 7.12–7.13. Comparison is made for two types of cables having the same ql/H ratio.

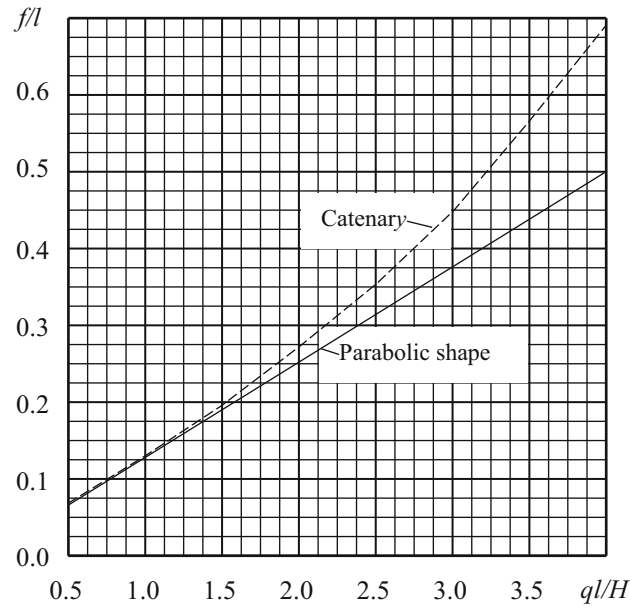


Fig. 7.12 Dimensionless sag-span f/l ratio vs. total load-thrust ql/H ratio for parabolic and catenary cables

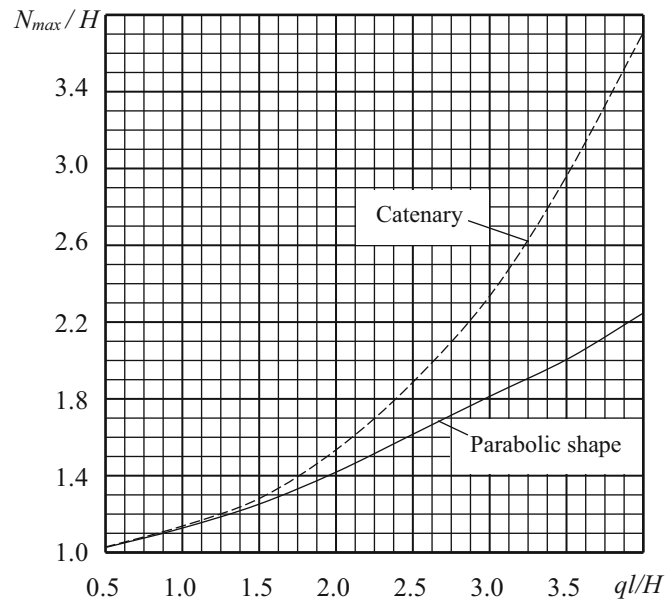


Fig. 7.13 Dimensionless maximum cable tension-thrust N_{max}/H ratio vs. total load-thrust ql/H ratio for parabolic and catenary cables

For thrust-shape problem the ratio ql/H is known. For relatively small ql/H (<1.5) dimensionless sag f/l and maximum tension N_{max}/H for parabolic and catenary cables practically coincide. If load q is fixed, then increasing of ql/H caused by decreasing of the thrust leads to increasing of the sag. If thrust H is fixed, then increasing of ql/H caused by increasing of the load q leads to increasing of the slope of the cable at support point, and as a result to increasing of maximum internal force N_{max} in cable.

7.6 Effect of Axial Stiffness

So far we have considered the behavior of a cable without accounting for elastic properties of a cable itself. Elastic properties of a cable have considerable importance in the distribution of tension of a cable and value of sag. Procedure of analysis of elastic cables is presented below. Two cases are considered: cable with concentrated and uniformly distributed loads.

7.6.1 Elastic Cable with Concentrated Load

Design diagram of elastic cable is presented in Fig. 7.14. The force P is applied at $x = l/2$. In the deformable state of the cable the sag and inclination are f and α , respectively. Non-deformable length of the cable is L_0 . It is necessary to calculate the thrust and sag of the cable taking into account the stiffness of the cable EA (E is modulus of elasticity; A is cross-sectional area of the cable).

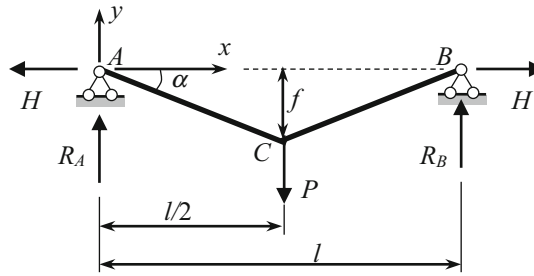


Fig. 7.14 Elastic cable subjected to concentrated load

It is evident that the angle of inclination and sag of the cable at point C are

$$\tan \alpha = -\frac{R_A}{H} = -\frac{P}{2H}$$

$$f = \frac{l}{2} \tan \alpha = \frac{Pl}{4H}$$

The total length of the cable is

$$L = 2 \frac{f}{\sin \alpha} = \frac{2}{\sin \alpha} \cdot \frac{l}{2} \tan \alpha = l \sqrt{1 + \tan^2 \alpha} = l \sqrt{1 + \frac{P^2}{4H^2}} \quad (7.37)$$

Tension in the left and right portions of a cable may be presented in the form

$$N = \frac{H}{\cos \alpha} = H \sqrt{1 + \tan^2 \alpha} = H \sqrt{1 + \frac{P^2}{4H^2}}$$

Now let us introduce the elastic properties of the cable; let the stiffness of the cable be EA . The strain of the cable is

$$\varepsilon = \frac{N}{EA} = \frac{H}{EA} \sqrt{1 + \frac{P^2}{4H^2}} \quad (7.38)$$

If the initial length of the cable is L_0 , then a length of the cable upon the load is

$$L = L_0(1 + \varepsilon) \quad (7.39)$$

Equations (7.37)–(7.39) lead to one resolving equation:

$$P = 2H \sqrt{\frac{\beta^2}{\left(1 - \frac{H}{EA} \cdot \beta\right)^2} - 1}, \quad \beta = \frac{L_0}{l} \quad (7.40)$$

This equation allows calculating unknown thrust H for given P , EA , initial length L_0 , and span l .
The sag of elastic cable is

$$f_{el} = \frac{Pl}{4H} = \frac{l}{2} \sqrt{\frac{\beta^2}{\left(1 - \frac{H}{EA} \beta\right)^2} - 1} \quad (7.41)$$

Limiting Cases

1. The cable is non-deformable ($EA = \infty$, $L_0 = L$). In this case, $f^2 + (l/2)^2 = (L/2)^2$, and expression for sag is

$$f_{nd} = \frac{l}{2} \sqrt{\frac{L^2}{l^2} - 1} \quad (7.41a)$$

Expressions (7.41) and (7.41a) allow evaluating the ratio of the sag for deformable and non-deformable cables. If $\beta = 1.5$ and $H = 0.1EA$, then $f_{el} = 1.22f_{nd}$.

2. The initial length of a cable equals the span ($L_0 = l$); it means that the cable may be treated as a string. In this case $\beta = 1$ and Eq. (7.40) becomes

$$P = 2H \sqrt{\frac{1}{\left(1 - \frac{H}{EA}\right)^2} - 1} \quad (7.42)$$

This equation after simple transformations may be rewritten in equivalent form

$$P = \frac{2\sqrt{2}}{1 - \frac{H}{EA}} \sqrt{\frac{H^3}{EA}} \sqrt{1 - \frac{H}{2EA}} \quad (7.42a)$$

Since $(H/EA) \ll 1$, Eq. (7.42a) may be presented as $P = \sqrt{\frac{8H^3}{EA}}$ or

$$H = \frac{1}{2} \cdot \sqrt[3]{P^2 EA} \quad (7.43)$$

Corresponding sag is

$$f = \frac{Pl}{4H} = \frac{l}{2} \cdot \sqrt[3]{\frac{P}{EA}} \quad (7.44)$$

Equations (7.43) and (7.44) show that the relationships P - H and P - f do not obey the linear law.

7.6.2 Elastic Cable with Uniformly Distributed Load

Now let us consider a gentle cable of the span l . In this case a distributed load may be considered as the sum of external load and the weight of a cable itself. Since the cable is gentle, we assume that the tension in the cable is constant and equal to the thrust

$$N = H \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \approx H$$

According to Eq. (7.16b) we have

$$\frac{L}{l} = 1 + \frac{1}{24} \frac{q^2 l^2}{H^2}.$$

Solution of this equation leads to the following expression for a thrust (in terms of span l , total length L of the cable, and load q):

$$H = \frac{ql}{2\sqrt{6}} \frac{1}{\sqrt{\frac{L}{l} - 1}} \quad (7.45)$$

The length L of the cable under load is $L = L_0(1 + \varepsilon)$, where L_0 is the initial length of the cable and $\varepsilon = \frac{H}{EA}$, so

$$L = L_0(1 + \varepsilon) = L_0 \left(1 + \frac{H}{EA}\right). \quad (7.46)$$

Substitution of (7.46) into (7.45) allows taking into account axial stiffness, so thrust H may be presented as

$$H = \frac{ql}{2\sqrt{6}} \cdot \frac{1}{\sqrt{\beta \left(1 + \frac{H}{EA}\right) - 1}}, \quad \beta = \frac{L_0}{l} \quad (7.47)$$

Equation (7.47) may be rewritten in the form of non-complete cubic equation with respect to trust H :

$$\frac{\beta}{EA} H^3 + (\beta - 1)H^2 = \frac{q^2 l^2}{24} \quad (7.47a)$$

Formula (7.10) for gentle cable $f = ql^2/8H$ and expression (7.47) for thrust lead to the following expression for sag of the gentle cable:

$$f = \frac{\sqrt{3}}{2\sqrt{2}} l \cdot \sqrt{\beta \left(1 + \frac{H}{EA}\right) - 1} \quad (7.48)$$

String. Assume that the initial length of the cable and span are equal ($L_0 = l$). In this case parameter $\beta = 1$. Corresponding solution of Eq. (7.47a) becomes

$$H = \sqrt[3]{\frac{q^2 l^2 EA}{24}} \quad (7.49)$$

$$\text{or } ql = 2\sqrt{6} \sqrt[3]{\frac{H^3}{EA}} \quad (7.49a)$$

Formulas (7.49a) and (7.47) for case $\beta = 1$ lead to the following nonlinear relationship f - q :

$$f = \frac{l}{4} \cdot \sqrt[3]{\frac{3ql}{EA}} \quad (7.50)$$

Limiting Case For non-deformable cable ($EA = \infty$), formulas (7.47) and (7.48) coincide with formulas (7.16c) and (7.16d) for case of gentle cable carrying uniformly distributed load.

We can see that consideration of the elastic properties of a cable leads to the nonlinear relationships H - q and f - q as well as H - EA and f - EA .

Problems

7.1. Design diagram of a cable is presented in Fig. P7.1. Supports A and B are located on different elevations. Parameters of the system are $a_1 = 10$ m, $a_2 = 22$ m, $c = 3$ m, $l = 30$ m, $P_1 = 18$ kN, and $P_2 = 15$ kN.

Determine the thrust H of the cable, if the total length of the cable is $L = 34$ m.

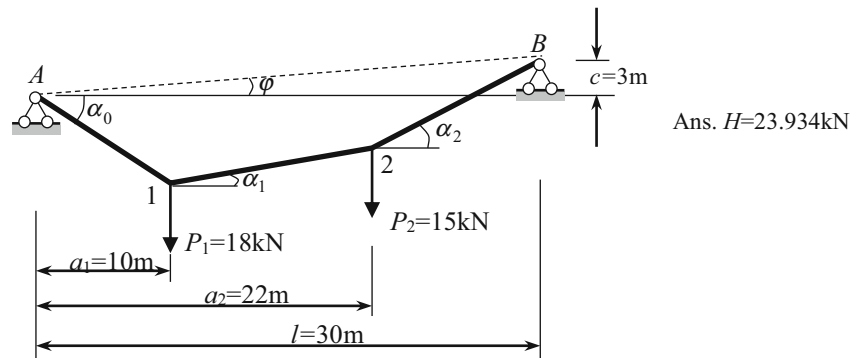


Fig. P7.1

7.2. The cable is subjected to two concentrated forces $P = 30$ kN at points E and C , and unknown force N at point D , as shown in Fig. P7.2. The thrust of the cable is $H = 60$ kN. Determine the force N and corresponding shape of the cable, if the portion CD of the cable is horizontal.

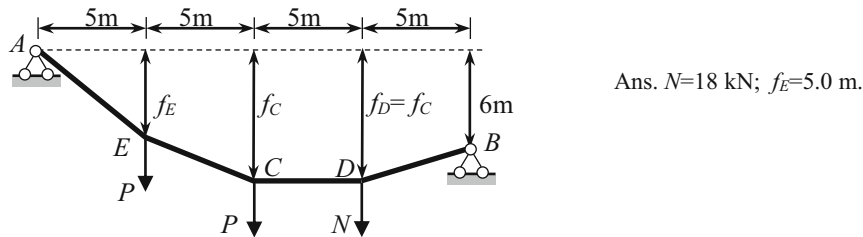


Fig. P7.2

7.3. The cable with support points on the same levels has span $l = 36$ m and is subjected to uniformly distributed load $q = 2$ kN/m along horizontal projection of the cable. The sag of the cable is $f = 6$ m. Determine the thrust of the cable H , maximum axial tension $N_{\max}$, and slope at the support. **Ans.** $H = 54$ kN; $N_{\max} = 64.89$ kN.

7.4. The cable with support points on the same levels has span $l = 36$ m and is subjected to uniformly distributed load $q = 2$ kN/m along horizontal projection of the cable. The thrust of the cable is $H = 108$ kN. Determine the sag of the cable, maximum axial tension $N_{\max}$, and slope at the support. **Ans.** $f = 3$ m; $N_{\max} = 113.84$ kN.

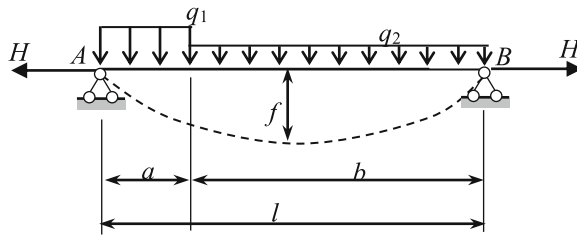
7.5. The cable with support points on the same elevation is subjected to uniformly distributed load q along the horizontal projection of the cable. The span of the cable is l . If load q increases two times, but the sag f remains the same, then

- Thrust H (1) remains the same; (2) is twice as much; and (3) is half as much.
- Maximum axial force (1) remains the same; (2) is twice as much; and (3) is half as much.
- Slope at the support (1) remains the same; (2) is twice as much; and (3) is half as much.

7.6. The cable with support points on the same levels is subjected to uniformly distributed load q along horizontal projection of the cable. The span of the cable is l . If load q increases two times, but the thrust H remains the same, then

- Sag of the cable (1) remains the same; (2) is greater; (3) is twice as much; and (4) is half as much.
- Maximum axial force (1) remains the same; (2) is greater; (3) is twice as much; and (4) is half as much.
- Slope at the support (1) remains the same; (2) is greater; (3) is twice as much; and (4) is half as much.

7.7. The flexible inextensible cable with support points on the same levels is subjected to uniformly distributed loads q_1 and q_2 within the horizontal line, as shown in Fig. P7.7; the span of the cable is l and maximum sag is f . Calculate the thrust of the cable. Consider limiting cases $q_1 = q_2$ and $q_2 = q_1$.



$$\text{Ans. } H = \frac{(q_1 a^2 + 2q_2 bl - q_2 b^2)^2}{8q_2 fl^2}$$

Fig. P7.7

7.8. The flexible inextensible cable with support points on the same elevation is subjected to uniformly distributed load q within the horizontal line; the span of the cable is l . Derive the expression for maximum tension $N_{\max}$ in terms of sag-to-span ratio $\alpha = fl$. Ans. $N_{\max} = \left(\frac{ql}{8}\right) \left(\frac{1}{\alpha}\right) \sqrt{1 + 16\alpha^2}$

7.9. The uniform cable of weight q_0 per unit length is suspended between two points at the same elevation and a distance l apart. Let a maximum axial force $N_{\max}$ and total weight of the cable W be related as $N_{\max} = kW$, where k is any positive number.

- Derive equation, which connects parameters q_0 , l , H , and k .
- Calculate the sag-span ratio for which the maximum tension in the cable is equal to 0.75 of the total weight of the entire cable and corresponding value of maximum tension $N_{\max}$.

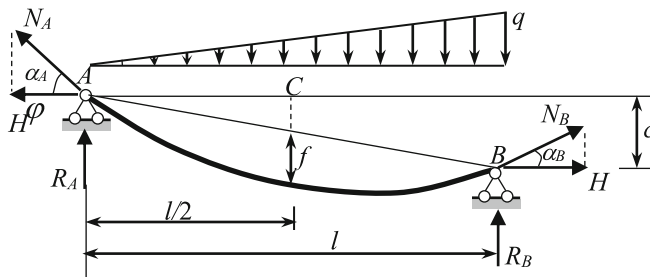
$$\text{Ans. (a). } (4k^2 - 1) \sinh^2 \frac{q_0 l}{2H} = 1; \text{ (b). } 0.2121, N_{\max} = 0.8337 q_0 l$$

7.10. The uniform cable of weight q_0 per unit length is suspended between two points at the same elevation and a distance l apart. Determine the sag-span ratio, for which the maximum tension is as small as possible. Ans. $fl = 0.3377$

7.11. The uniform cable of weight q_0 per unit length is suspended between two points at the same elevation and a distance l apart. The sag of the cable, total length, and thrust are denoted as f , L , and H , respectively. Calculate the maximum tensile force. Present the result in three following forms: (1) in terms of H , q_0 , and l ; (2) in terms of H , q_0 , and f ; and (3) in terms of H , q_0 , and L .

$$\text{Ans. (1) } N_{\max} = H \cosh \frac{q_0 l}{2H}; \text{ (2) } N_{\max} = q_0 f + H; \text{ (3) } N_{\max} = H \sqrt{1 + \left(\frac{q_0 L}{2H}\right)^2}$$

7.12. Design diagram of flexible cable with support points A and B on different levels is presented in Fig. P7.12. The cable is subjected to linearly distributed load q . At the middle point C the sag is $f = 4.5$ m. Find the shape of the cable and thrust and calculate the distribution of internal forces. Parameters of the system are $l = 60$ m, $c = 12$ m, and $q = 2.0$ kN/m. Use the concept of the reference beam.



$$\begin{aligned} \text{Ans. } H &= 100 \text{ kN}, \\ N_A &= 107.76 \text{ kN}, \\ N_B &= 101.83 \text{ kN}, \\ R_A &= 40 \text{ kN}, \quad R_B = 20 \text{ kN} \end{aligned}$$

Fig. P7.12